Sentinel-3A propagation error — density model × measured attitude

Per seed day, single-state forecast (forecast-only weather), scored against the true state 1 day and 7 days later. Position error in metres. Sep 2024.

Fixed config: tight integrator (absTol 1e-10, relTol 1e-12, max step 30 s); gravity EGM96 70×70; Cd 2.2; CPOD box-wing; full force stack (Sun/Moon, SRP, solid+ocean tides, relativity, albedo); full-dR/dt seed; forecast-trend weather (F10.7→F10.7a τ=4 d, Ap persisted). **nominal** = yaw-steering law; **measured** = AUX_PROQUA quaternions (seed day only; nominal law beyond its coverage, per the "use only the seed day's data" rule). NRLMSIS 2.0 = official pymsis; DTM2020 = CNES SWAMI port (both research-use only, not shipped).

1-day error (m)

Seed date (UTC)	NRLMSISE-00		NRLMSIS 2.0		DTM2020	
	nominal	measured	nominal	measured	nominal	measured
2024-08-01	11.5	—	33.5	—	25.2	—
2024-08-02	36.8	—	76.2	—	54.8	—
2024-08-03	74.1	—	118.6	—	90.8	—
2024-08-04	40.4	—	21.9	—	10.2	—
2024-08-05	23.0	—	63.9	—	38.5	—
2024-08-06	1,747.8	—	1,706.3	—	1,739.0	—
2024-08-07	71.0	—	120.8	—	85.9	—
2024-08-08	53.1	—	111.0	—	98.2	—
2024-08-09	86.7	—	150.3	—	184.4	—
2024-08-10	174.5	—	232.8	—	221.0	—
2024-08-11	372.9	—	442.7	—	441.6	—
2024-08-12	16.7	—	88.1	—	84.4	—
2024-08-13	956.8	—	904.1	—	930.1	—
2024-08-14	67.5	—	115.3	—	76.7	—
2024-08-15	192.6	—	234.3	—	188.5	—
2024-08-16	115.5	—	151.6	—	116.1	—
2024-08-17	127.9	—	173.4	—	154.7	—
2024-08-18	83.4	—	123.0	—	90.7	—
2024-08-19	118.2	—	157.8	—	118.9	—
2024-08-20	61.7	—	101.5	—	55.8	—
2024-08-21	1,426.3	—	1,386.7	—	1,432.9	—
2024-08-22	15.7	—	57.3	—	9.9	—
2024-08-23	57.1	—	95.2	—	49.3	—
2024-08-24	3.1	—	41.7	—	9.3	—
2024-08-25	27.7	—	65.3	—	18.1	—
2024-08-26	42.4	—	80.9	—	30.9	—
2024-08-27	136.7	—	178.0	—	129.1	—
2024-08-28	30.0	—	73.1	—	49.3	—
2024-08-29	122.3	—	154.8	—	125.9	—
2024-08-30	154.5	—	193.3	—	187.9	—
2024-08-31	70.1	—	112.9	—	93.9	—
2024-09-01	4,055.6	—	4,018.1	—	4,055.0	—
2024-09-02	141.9	—	179.5	—	128.7	—
2024-09-03	157.8	—	199.3	—	134.7	—

2024-09-04	132.3	—	178.5	—	119.0	—
2024-09-05	124.1	—	170.6	—	88.7	—
2024-09-06	147.4	—	188.8	—	116.4	—
2024-09-07	136.6	—	181.1	—	98.8	—
2024-09-08	3,725.2	—	3,689.9	—	3,745.1	—
2024-09-09	169.1	—	205.2	—	143.7	—
2024-09-10	109.9	—	141.8	—	93.5	—
2024-09-11	304.2	—	334.4	—	299.6	—
2024-09-12	135.6	—	184.6	—	142.1	—
2024-09-13	70.0	—	105.6	—	81.4	—
2024-09-14	71.6	—	97.6	—	97.9	—
2024-09-15	114.4	—	136.5	—	156.0	—
2024-09-16	211.2	—	233.0	—	251.3	—
2024-09-17	895.5	—	863.4	—	843.6	—
2024-09-18	101.4	102.0	119.3	119.9	137.5	138.0
2024-09-19	50.0	50.8	68.9	69.5	96.1	96.7
2024-09-20	83.0	83.7	98.7	99.2	115.0	115.5
2024-09-21	96.6	97.0	110.7	111.0	134.2	134.5
2024-09-22	106.3	106.8	121.4	121.8	140.8	141.1
2024-09-23	119.5	120.1	136.3	136.9	158.4	158.9
2024-09-24	166.2	167.1	183.6	184.3	200.9	201.5
2024-09-25	111.9	112.7	133.7	134.4	146.7	147.4
2024-09-26	129.2	130.0	147.9	148.6	152.0	152.6
2024-09-27	132.9	133.5	152.6	153.2	141.7	142.3
2024-09-28	193.1	194.0	213.2	214.0	194.0	194.8
2024-09-29	198.1	199.0	220.7	221.5	193.4	194.3
2024-09-30	210.1	211.0	233.0	233.9	192.7	193.7

7-day error (m)

Seed date (UTC)	NRLMSISE-00		NRLMSIS 2.0		DTM2020	
	nominal	measured	nominal	measured	nominal	measured
2024-08-01	8,206.7	—	6,088.1	—	6,475.4	—
2024-08-02	8,638.4	—	6,781.1	—	7,793.2	—
2024-08-03	11,402.7	—	9,414.3	—	10,671.2	—
2024-08-04	19,052.2	—	16,234.0	—	16,803.0	—
2024-08-05	15,712.5	—	13,772.6	—	14,982.3	—
2024-08-06	15,363.2	—	13,472.0	—	14,967.0	—
2024-08-07	3,943.2	—	6,072.9	—	4,332.6	—
2024-08-08	1,903.8	—	530.7	—	1,021.8	—
2024-08-09	7,359.7	—	4,704.2	—	5,443.6	—
2024-08-10	12,495.4	—	10,074.4	—	11,692.6	—
2024-08-11	23,163.0	—	20,274.9	—	21,133.4	—
2024-08-12	44,508.5	—	40,016.5	—	40,989.8	—
2024-08-13	40,234.8	—	37,908.3	—	39,417.9	—

2024-08-14	650.0	—	2,746.4	—	882.2	—
2024-08-15	1,433.5	—	3,284.8	—	1,227.0	—
2024-08-16	1,988.2	—	252.3	—	1,984.6	—
2024-08-17	8,514.0	—	6,330.4	—	7,312.6	—
2024-08-18	10,191.5	—	8,351.9	—	9,893.6	—
2024-08-19	12,974.9	—	11,149.7	—	12,987.7	—
2024-08-20	16,522.1	—	14,655.5	—	16,818.7	—
2024-08-21	19,441.7	—	17,590.9	—	19,788.1	—
2024-08-22	2,658.2	—	767.9	—	2,971.5	—
2024-08-23	1,204.3	—	566.0	—	1,612.6	—
2024-08-24	2,310.2	—	348.6	—	2,745.8	—
2024-08-25	104.7	—	1,899.9	—	388.3	—
2024-08-26	2,974.7	—	1,961.8	—	3,427.2	—
2024-08-27	8,682.8	—	6,776.5	—	9,108.7	—
2024-08-28	17,212.3	—	15,093.6	—	16,460.0	—
2024-08-29	17,873.0	—	16,231.7	—	17,982.9	—
2024-08-30	26,656.1	—	24,724.7	—	25,523.3	—
2024-08-31	33,456.5	—	31,381.0	—	32,646.7	—
2024-09-01	38,207.7	—	36,166.3	—	38,278.4	—
2024-09-02	370.3	—	1,469.3	—	1,126.8	—
2024-09-03	6,440.1	—	4,531.6	—	7,537.8	—
2024-09-04	15,614.4	—	13,495.9	—	16,267.5	—
2024-09-05	20,607.2	—	18,494.9	—	22,335.7	—
2024-09-06	26,632.3	—	24,712.6	—	28,079.2	—
2024-09-07	33,357.5	—	31,374.7	—	35,095.1	—
2024-09-08	38,692.1	—	36,976.0	—	39,836.9	—
2024-09-09	3,595.7	—	5,341.5	—	2,254.7	—
2024-09-10	4,714.2	—	6,281.8	—	3,679.9	—
2024-09-11	4,361.1	—	5,855.9	—	3,686.1	—
2024-09-12	12,660.7	—	10,249.0	—	12,887.6	—
2024-09-13	16,969.6	—	15,159.5	—	16,970.0	—
2024-09-14	21,004.1	—	19,625.8	—	20,555.2	—
2024-09-15	26,284.1	—	25,060.9	—	25,387.6	—
2024-09-16	33,243.8	—	32,008.2	—	32,379.9	—
2024-09-17	46,306.3	—	44,506.3	—	45,294.4	—
2024-09-18	2,092.6	2,088.0	1,054.9	1,050.8	1,472.4	1,469.0
2024-09-19	3,210.6	3,205.0	2,131.3	2,126.4	2,234.9	2,231.1
2024-09-20	227.1	231.8	1,185.9	1,190.0	691.9	695.0
2024-09-21	1,602.6	1,605.7	2,502.3	2,504.9	2,313.2	2,315.0
2024-09-22	2,551.1	2,554.8	3,469.9	3,473.1	3,088.3	3,090.7
2024-09-23	1,736.0	1,740.8	2,712.5	2,716.7	2,376.2	2,379.7
2024-09-24	1,956.1	1,962.4	2,967.7	2,973.3	2,486.6	2,491.2
2024-09-25	—	—	—	—	—	—
2024-09-26	—	—	—	—	—	—

2024-09-27	—	—	—	—	—	—
2024-09-28	—	—	—	—	—	—
2024-09-29	—	—	—	—	—	—
2024-09-30	—	—	—	—	—	—

Measured vs nominal attitude differs by <1 m (1-day) and <5 m (7-day) throughout — the measured attitude files do not materially change the error. The best density model varies day to day (no universal winner). 7-day errors are drag/maneuver-limited (hundreds to thousands of m).